



An IMU-based traffic and road condition monitoring system

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ABSTRACT

This paper presents a new type of wireless platform designed for real-time traffic estimation and road surface monitoring. The sensor platform is built around a 32-bit ARM Cortex M4 micro-controller and a LSM9DS0 Inertial Measurement Unit module. This platform is mainly designed for probe vehicles and can be easily installed in a vehicle equipped with a USB charger. The hardware architecture design and software programming system of the proposed platform are introduced as well as its cost evaluation from the first generation to the third generation. The article then demonstrates some applications of such a platform in smart cities, including trajectory estimation and road condition monitoring. All design files have been uploaded and shared in an open science framework, and can be accessed from https://osf.io/524y9/?view_only=85859a345a7b429fbb8fe194966daa5b. It is licensed under the GNU General Public License v3.0.

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1. Introduction

1.1. Traffic estimation and road condition monitoring

The objective of real-time traffic estimation is to estimate traffic related variables (e.g. location, speed, flow) based on measurement data in order to obtain a complete view of the traffic situation [1]. Traffic estimation allows one to generate traffic maps [2], estimate travel time [3], optimal routes for vehicles, or optimal control policies [4] for traffic control systems [5]. All these applications require the estimation of the current and near future traffic state across the road network, using real-time data [6]. Traffic estimation requires the availability of real-time data, which is usually generated through traffic monitoring systems. Most legacy traffic sensing systems are based on fixed sensors, which are expensive to deploy and maintain. Mobile sensors are usually based on GPS or satellite-based positioning systems, and have to be in range of satellites. The later system is less expensive than fixed sensors, though its accuracy in urban area is reduced due to obstacles and the urban canyon effect [7–9]. Another important application of traffic estimation systems is road condition monitoring, which aims at detecting any surface imperfections (such as road surface cracks) for paved or unpaved roads [10] at early stages. Road condition monitoring plays a key role in ensuring safety and comfort to various road users, from pedestrians to drivers. Furthermore, having information on infrastructure quality guarantees road managers an adequate maintenance on time [11,12]. Previous vehicle-equipped road condition monitoring technologies based on IMUs, laser scanners, road profilers, image and video cameras are usually associated with high purchase, operation or maintenance costs [13]. In order to be more cost-effective, a low cost and high accuracy IMU-based monitoring system that can be applied for both applications is developed. Since IMUs do not generate absolute position measurement data needed for traffic estimation or direct

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information of indicating road surface condition, some essential computational tasks should be implemented to capture such features. Therefore, the development of a new platform and software capable of meeting these specifications is required.

1.2. Lagrangian sensing

Traffic monitoring system is generally categorized as fixed (Eulerian) or mobile/probe (Lagrangian) based system. The former often includes a variety of sensors such as radars, inductive loop detectors and traffic cameras [14–16], which are installed at a fixed point. In contrast, a Lagrangian-based system, such as GPS, relies on the data generated by vehicle, and measures traffic conditions along the path of the vehicle. In the past decades, Lagrangian sensing methods become increasingly prevalent in modern traffic monitoring systems.

As one of the Lagrangian sensing methods, vehicle probe technology is emerging as a viable means for traffic monitoring, delivering speed, location and time information for the purposes of managing the transportation network [17,18]. While probe data is relatively accurate and has an extremely low marginal cost, many issues still remain associated with this technology, particularly the low penetration rate of participating users due to a lack of interest and a lack of user privacy guarantees [19,20]. For instance, a GPS-based system needs the users to share their location data. Other issues such as high power consumption (GPS in cellphones) or higher production cost also prevent the large-scale deployment of such systems. Keeping all these considerations in mind, it was decided to develop a low-cost, low-power Lagrangian system applicable for probe vehicles, which can guarantee user privacy since it would ultimately operate without an absolute positioning device.

1.3. Wireless sensor networks

Wireless Sensor Networks (WSNs) have emerged as an essential solution to urban monitoring applications because of their computation, communication and sensing capabilities. For most of the urban sensing systems, the cost of deployment and maintenance is usually much higher than the cost of hardware. Compared with these systems, WSNs feature easier deployment and better flexibility. With this great advancement, WSNs have been used for countless applications in many different fields such as environmental monitoring, structural health monitoring and seismic activity detection [21–23]. In this case, the proposed system requires to be fully wireless so that it can be flexible enough for probe vehicles and at the same time minimize the deployment cost.

However, there are some remaining limitations associated with the operation of WSNs, for instance, energy supply, communications range/bandwidth, computation, and storage. Even though significant work has been done to reduce power consumption for WSNs systems, their computational capabilities are usually limited. Many systems do not run complex algorithms on board to ensure low power consumption and real-time operation. For example, [24] runs a 40 MHz operation MPC555 Microcontroller, while most of the complex computation tasks are handled at computer server levels. Compared with such systems, the proposed system requires to conduct complex algorithms on board such as the auto-calibration of the IMU, and the trajectory reconstruction from IMU data. This procedure involves a complex optimization problem related to the accelerometer measurements. Based on all these considerations, a low cost and high computational performance platform is required.

2. Computational requirements

Even though the low-cost IMU can generate high frequency and accurate sensing data, it does not generate speed or positioning data directly, which is necessary for most of the traffic-related applications. This article relies on the previous article [25] on IMUs for the sensing principle, and complements the latter with road condition monitoring. For these tasks, an effective computational platform is needed to generate the measurement data. Also, each node in this system has to carry out a number of computational tasks for real-time traffic and roadway monitoring:

- Automatic calibration of the IMU including the calculation of a 3×3 rotation matrix based on linear fitting algorithm [25]
- Trajectory estimation, including the least square optimization algorithm for speed estimation and the DCM filter for attitude estimation [25]
- Road surface condition monitoring based on IMU data, including a linear regression algorithm for monitoring and prediction of road surface condition index.

3. Platform architecture and design

To conduct the aforementioned computational tasks in real time, a hardware platform was designed to simplify programming. Following these requirements, the proposed platform should have some specific features in comparison with other reported hardware platforms.

1. Low node and deployment cost, low operation power;
2. Small size and straightforward installation for probe vehicles;

- Advanced computational capabilities, with enough free memory to allocate relatively large matrices required for self-calibration, speed and trajectory estimation, and other computational tasks needed for the application.

Fig. 1 shows the block diagram of the hardware platform. A detailed description of this hardware platform was given by focusing on the following areas: the processing unit, communication, data storage, and peripherals in the following part. Fig. 2 illustrates the third version of the proposed hardware platform.

3.1. Microcontroller

The core component of the sensing platform is the Microcontroller (MCU), which handles sensing (Analog-to-digital Converter and digital buses), computation and control. The STM32F407, a 32-bit ARM Cortex-M4 based microcontroller from ST is selected since it balances the trade-off among computation, RAM, power consumption and cost very well. A wide range of microcontrollers are considered, in which the ATmega1281 is on the low end (low performance, low power consumption) and the TI TMS570 on the high end (high performance, high power consumption). The MCUs in the low end are not able to provide sufficient internal data RAM (8 Mb), program memory (128 kb) and computational power (16 MHz). On the other hand, the high end exhibits a fast frequency (180 MHz). However, they also have higher power consumption and higher prices, which makes them unsuitable for mass scale production. In contrast, the STM32F407 provides a comparable performance with TMS570 for only one third of the price. In addition, at its lowest frequency setting its power consumption is comparable to the power consumption of the ATmega1281.

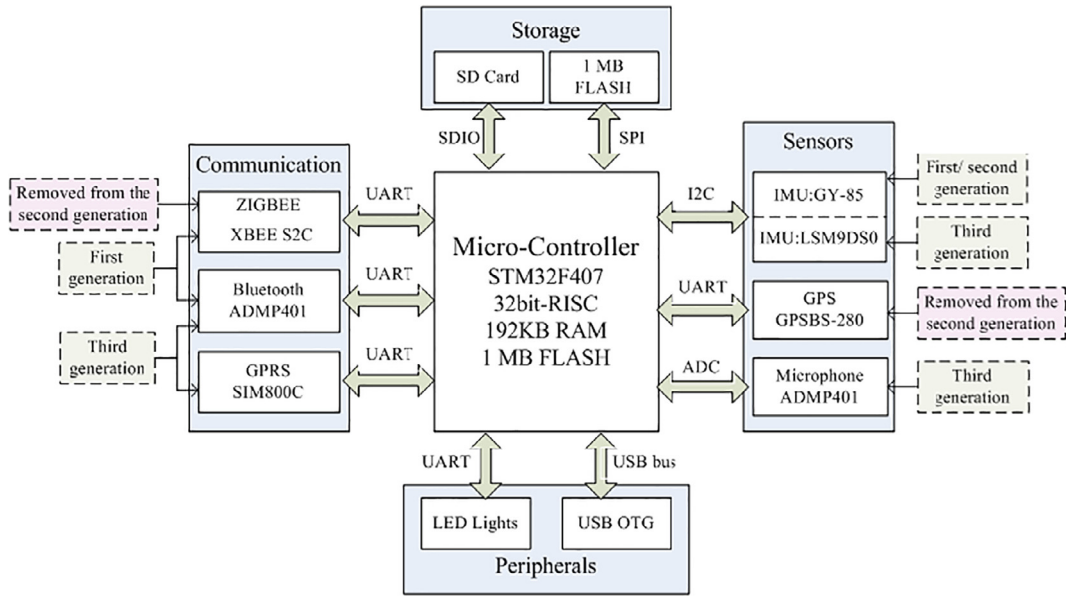


Fig. 1. Block diagram of the IMU-based platform.

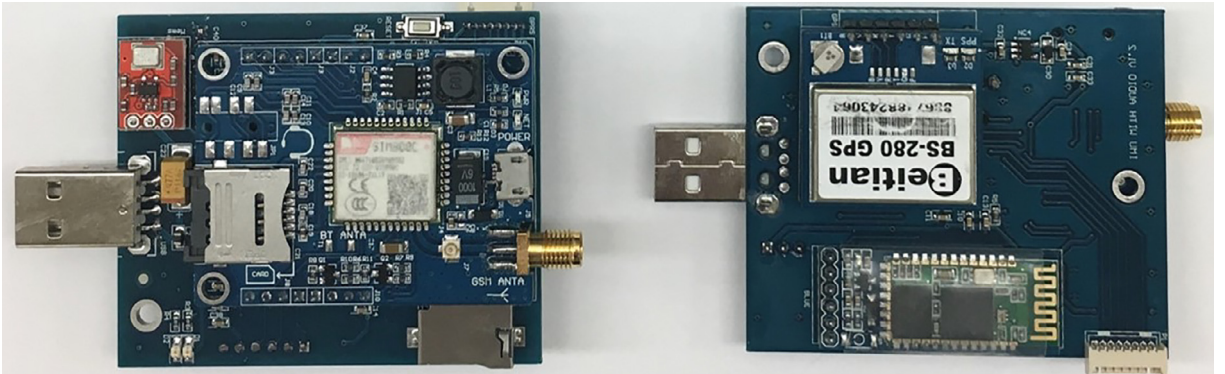


Fig. 2. Hardware platform of the third generation.

The STM32F407 includes a 1 MB Flash memory and 196 KB of data RAM. It supports up to seventeen timers, a 24 channels Analog-to-digital Converter (ADC) and two 12-bit Digital-to-Analog Converters (DACs) for peripherals. On this proposed platform, the microcontroller is configured to have three Universal Asynchronous Receivers/Transmitters (UARTs) for communication and positioning modules, one I2C interface bus and one ADC interface for sensors. Furthermore, a Secure Digital Input Output (SDIO) and a USB OTG (On-The-Go) bus are configured to provide Micro SD Flash storage and USB host access. The STM32F407 supports a maximum frequency of 168 MHz, which is sufficient to run the envisioned traffic sensing and estimation algorithms in real time.

3.2. Communications

The transmission of data between different sensor nodes requires the use of radio transceiver. For the first generation, XBEE S2C was chosen from Digi working at 2.4 GHz, using the IEEE 802.15.4 standard. This transceiver is capable of generating signals up to +18 dBm, which is the maximal legally allowed transmission power in the 2.4 GHz band (equivalent to 100 mW EIRP when combined with a 2 dB dipole antenna). There are a number of 802.15.4 compliant radio transceivers available, such as the TI CC2500, though their maximal radiated power is insufficient for our application.

In addition, a Bluetooth module is used for dynamic data transfer from the sensor to the mobile device. The selected module is the SH-HC-06, which has the Blue Core4-Ext chip, and follows the Bluetooth V2.0 + EDR Bluetooth standard. The maximum transfer rate of this transceiver is 2.1 Mb/s, and the transmission distance is more than 20 m. Compared with other possible choices, this module has low cost, small size and a high sensitivity which is up to -80dBm at 0.1% BER (bit error rate). It supports AT commands to modify serial baud rate, device name, pairing passwords and other parameters. As for the software, a mobile client was developed with an Android operating system for data transmission and storage. Fig. 3 shows the data transmission through Bluetooth. The primary wireless communication used in this article is Bluetooth, with traffic Bluetooth readers acting as fixed nodes in the sensor network.

The GPRS SIM800C module is used in the third generation platform, for remote monitoring purpose and communicating the sensing data to remote server via the cellular phone network. This transceiver is selected mainly for its low power consumption and small size. The SIM800C is designed with power saving technique, and the current consumption can be as low as 0.6 mA while in sleep mode. It has a tiny configuration of only $17.6 \times 15.7 \times 2.3$ mm, which can meet the space saving requirement for our platform. This module supports 4G working on frequencies GSM850MHz, EGSM900MHz, DCS1800MHz and PCS1900MHz. A micro SIM card is used in this transceiver for data storage and communication. Based on these communication modules, a WSN can be built. Data generated by the sensing device can be sent to fixed nodes for processing, either through Bluetooth or through the cellular network.

3.3. Data storage

The MCU has an internal 1 MB FLASH memory for storing the bootloader, firmware, the operating system and a 196 KB internal SRAM for data during firmware execution. While this amount of memory is sufficient for real-time processing to perform the necessary computations, additional storage for non-volatile data (for instance measurement data needed for trajectory estimation, historical acceleration from IMU and audio spectrum information from Audio Processing module needed



Fig. 3. Transmission of data to an Android device via Bluetooth.

for pavement condition monitoring) is still needed. Thus a micro SD slot (Micro SD Flash Socket) is added, which is accessed through a SDIO interface.

3.4. Sensors

The main sensor that is incorporated into the embedded system for traffic estimation is the IMU sensor. IMUs are based on a combination of accelerometers, gyroscopes and magnetometers, which can be used to determine the accelerations and rotation rates of a vehicle, and its orientation using magnetic field measurements. IMUs do not require any external infrastructure to work and do not receive or transmit data wirelessly. They require an extremely low power to operate, considerably less than GPS or cellphone-based systems. Owing to their much lower complexity than GPS systems, IMUs are less expensive to manufacture than the latter [24,26]. They do not require an antenna for receiving signals, and are not at risk of losing connectivity with positioning satellites, which frequently happens with GPS systems, particularly if obstructions are present between the receiver and the satellites. Because of their high accuracy (over short time windows), IMUs are good at detecting and classifying the type of congestion encountered (traffic light, stop and go waves, slow and continuous traffic) [27,28]. In addition, such a system offers strong guarantee for the privacy of the participating users when used in conjunction with a short-range wireless sensor network [29,30]. Those features of IMUs make it more reliable and potentially more accurate than a GPS based positioning system for traffic measurement purposes.

In the proposed platform, IMU GY-85 is used for the first and second generation, which consists of ITG3205, ADXL345 and HMC5883 chips. It is replaced by LSM9DS0 for the third generation for higher accuracy considering the application of road surface monitoring. The LSM9DS0 is a system-in-package featuring a 3D digital linear acceleration sensor, a 3D digital angular rate sensor, and a 3D digital magnetic sensor. The embedded self-test allows the linear acceleration sensor functionality to be tested without moving. This module is connected to the MCU with I2C serial interface and provides a 16-bit data output.

The IMU is the main sensor used in our platform to obtain real time traffic data for traffic estimation and other applications, for instance, in the context of trajectory estimation. In order to verify the accuracy of the estimated results by IMU, the exact location of the vehicle is required as validation. Thus, a GPS receiver is used in our platform for getting vehicles localization. It is required for software development and validation only. For this purpose, the Beitian BS-280 GPS is selected in our platform for its relatively high output frequency with a low price and low power requirement. The module is integrated with the UBLOX G7020-KT chip, and can reach an output frequency of up to 10 Hz. Through a proprietary binary protocol, it provides position and velocity information at a 1 Hz rate. Also, it is designed with a tiny size of 28 mm × 28 mm × 10 mm, which is also suitable for the proposed platform.

An important application target of this proposed platform is to detect pavement condition, one necessary step is to estimate and quantify the audio noise which could be a predictor of the road condition. Thus a digital MEMS microphone is embedded in our platform for detecting ambient noise and the rolling noise component. One important computational task needs to be conducted for this system is to sample the sensing signals and divide them into different frequency bands through FFT (Fast Fourier Transform), in order to identify the rolling noise component and evaluate its properties. This step can be achieved through the DSP (Digital Signal Processing) library of the MCU (STM32F407). The audio acquisition module selected for our platform is the ADMP401 omnidirectional microphone, which is connected to an ADC of the MCU. The 12-bit ADC of STM32F407 (MCU) could reach the highest sample rate of 1 MHz (when the clock of CPU reaches 56 MHz). The ADMP401 microphone is designed as a 4.72 mm × 3.76 mm × 1.0 mm surface-mount package that can meet our space constraints. Besides, the SNR (Signal to Noise Ratio) of ADMP401 can be as high as 60 dBA and the sensitivity is of -42 dBA with a relatively low price. It also has a flat wideband frequency response from 100 Hz to 15 kHz, the expected frequency bands of rolling noise. What's more, its current consumption is extremely low (as listed in Table 1).

3.5. Other embedded auxiliary equipment/Peripherals

The peripherals consist of several functioning blocks: LEDs, a self-resetting circuitry and a USB monitoring unit.

As the proposed platform works with high modularity, different modules play different roles (sensing, communicating or storage). It is also important to make sure each part is working well to support the overall functioning of the whole system. Thus several LED lights were embedded in our platform to indicate the normal functioning of each module in case of running error. In the first generation, three LED lights were embedded in our platform to indicate the status of Bluetooth, GPS and IMU modules.

Software bugs are an important risk for every firmware, particularly in embedded systems. To anticipate the presence of software bugs, a self-resetting circuitry was included in our platform to prevent complete system failures. The functionality of the circuitry is to reset the whole system while any part of it is not functioning normally. The reset circuitry is based on a reset chip (MAX6468US16D2-T), when it is detected that the MCU supply voltage is lower than the reset chip voltage value, the MCU will reset. Besides, a reset button is designed on the platform for conveniently resetting the whole system through reset pin.

This platform is mainly designed for probe vehicles, and needed to be rigidly mounted in the vehicle, and continuously powered by an external supply. To achieve this, we added a USB port to the device. The USB port is used solely for charging

Table 1

Cost of the major components (excludes manufacturing costs).

Version	Item	Quantity	Price \$	Breakdown price \$	Remarks
First generation	STM32F407	1	11.05	7.18@1000	Micro-controller
	SH-HC-06	1	8.99	5.46@500	Bluetooth Transceiver
	Beitian BS-280	1	12.19	10.79@100	GPS
	XBP24CZ7UIT-004	1	10.40	8.53@500	XBEE Transceiver
	GY-85	1	8.45	5.82@300	IMU Sensor
	Total	1	50.72	37.38@1000	
Second generation	STM32F407	1	11.05	7.18@1000	Micro-controller
	SH-HC-06	1	8.99	5.46@500	Bluetooth Transceiver
	XBP24CZ7UIT-004	1	10.40	8.53@500	XBEE Transceiver
	GY-85	1	8.45	5.82@300	IMU Sensor
	Total	1	38.89	26.99@1000	
Third generation	STM32F407	1	11.05	7.18@1000	Micro-controller
	SH-HC-06	1	8.99	5.46@500	Bluetooth Transceiver
	SIM800C	1	8.75	5@500	GPRS Transceiver
	Beitian BS-280	1	12.19	10.79@100	GPS
	LSM9DS0	1	7.11	3.83@3000	IMU Sensor
	ADMP401	1	6.99	4.12@500	Microphone
	Total	1	55.08	36.39@3000	

the device (through a USB car charger), and also plays the role of a structural support that maintains the device in a constant (albeit unknown) orientation with respect to the vehicle frame, despite the accelerations, turns, and road bumps.

4. Software

The software used for the programming of the presented embedded system is Keil MDK. The μ Vision IDE used in this system provides an efficient environment that can create software applications using pre-built software components and device support from Software Packs. The ARM Compiler incorporates a highly optimized C/C++ compiler, assembler, linker and libraries for embedded software development. It involves powerful optimization techniques such as loop unrolling and function inlining, which are necessary for our platform to run these complex computational tasks such as the auto-calibration of the IMU sensor. Also, it helps make a good management between different modules. The programming interface and cable connection while loading the code from the programming system to the hardware are shown in Fig. 4.

For most embedded systems, data is collected from sensors at periodic times (multiples of the time step) and little care is given to the time delays between the various sensors. Neglecting these delays may result in inaccuracies for real-time algorithms [24]. For the proposed platform, accurately time-stamped data is extremely important for running efficient real-time traffic estimation and road condition monitoring algorithms on board. Our solution is to set different time conditions for different sensors to coordinate the sampling rate of the whole system. In this system, the IMU used in the platform can reach a high sampling rate of 200 Hz, while the sampling rate of the GPS sensor is only 1 Hz. Also, compared with IMU, reading data from GPS takes longer time, which will cause time delay and influence the running speed for the whole system. Thus

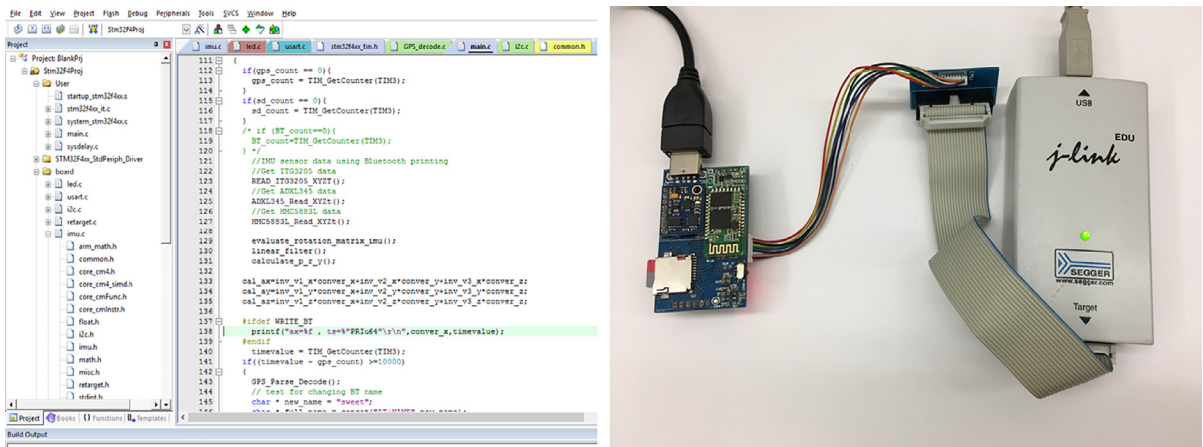


Fig. 4. The programming interface (left) and the cable connection while loading (right).

different time intervals are set for reading data from different sensors based on their highest sampling rates, and then store the data at a fixed sample rate. This process is conducted through setting the same timer but different conditions for the measurement of different sensors. In practice, we achieved the sample rate of 10 Hz when running all trajectory estimation algorithms, which is sufficient given the time scales associated with driving.

5. Platform cost evaluation

The costs and functions of the major components used for the proposed platform are listed in Table 1. The entire cost of the subsystems of the third version device is around \$55 including all the sensing, communication and storage modules. The proposed probe vehicle device, which cost around \$40, though the cost would be further reduced for high production volumes.

6. Applications of the platform

6.1. Trajectory estimation

One important application of the proposed platform is to estimate the trajectory of a vehicle using the measurements of the IMUs. The sensing platform is connected to a moving vehicle with a fixed angle (depends on how the USB interface in the vehicle is designed), thus the first step we need to take is to map the coordinates of the sensor to the coordinates of the vehicle, which is referred to automatic calibration and has been illustrated in [25]. Once the IMU device is calibrated, the trajectory of the probe vehicle can be estimated through measurements of accelerations, rotation rates and magnetic field along its path.

6.1.1. Least squares optimization for speed estimation considering sensor bias

Based on kinematic relation, speed can be obtained from the integration of acceleration, thus the discrete measurement of vehicle's longitudinal acceleration (a_x) can be used for estimating the speed through taking Riemann sum. In order to reduce calculation error caused by sensor noise and integration error, more steps need to be taken for estimating sensor bias and providing velocity measurements at certain positions for validation. The measured velocity can be obtained through detecting the stops (where velocity is zero) and turns (where the velocity can be estimated through the kinematic relation between lateral acceleration a_y and the rotation rate around the vertical axis g_z) along the path. Then the accelerometer bias can be estimated through solving an unconstrained least squares optimization problem, which is illustrated in Eq. (1).

$$\min_{d_1 \dots d_p} \sum_{i \in M} \left(v_0 + \left(\sum_{k=0}^p [a_x(k\Delta t) - d_m] \Delta t \right) - \mathbf{v}_{measured}(i\Delta t) \right)^2 \quad (1)$$

In Eq. (1), $d_1 \dots d_p$ refers to piecewise constant and unknown acceleration bias where the drifts are a multiple m of our integration step Δt . M corresponds to the set of integer time instants for which velocity measurements are available (at stops or turns). And the measurement velocity $\mathbf{v}_{measured}$ is obtained through stop (where the rotation rates are almost zero and the acceleration is almost constant on all axes) and turn (where the value of vehicle's rotation rate around the vertical axis becomes nonzero) detection, which is also illustrated in [25]. Results of speed estimation examples are shown in Fig. 5.

As can be seen from the figure, the estimated speed considering the acceleration bias is much closer to vehicle speed obtained from GPS (green color), while the directly integrated speed diverges to a very low (unrealistic) value. Also, it is

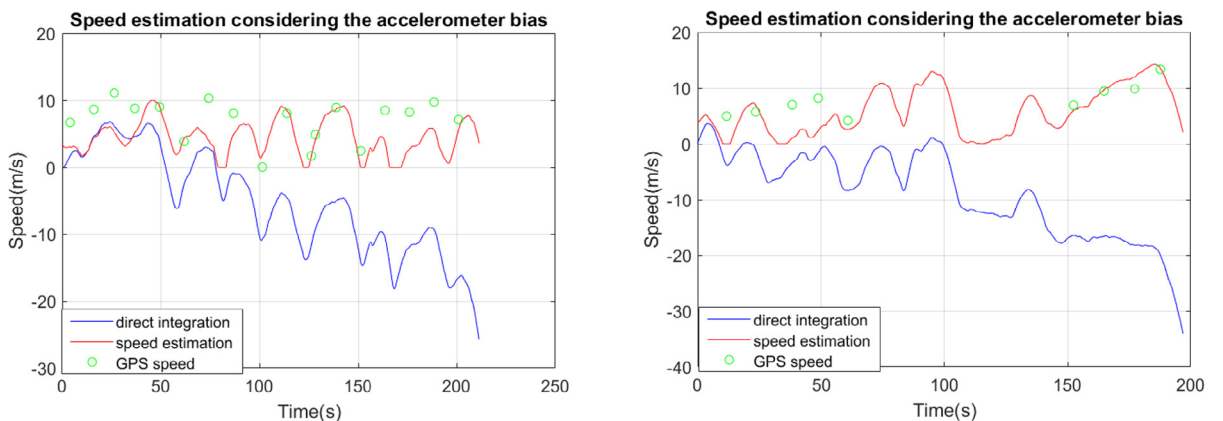


Fig. 5. Speed estimation results considering sensor bias in real tests.

shown in this figure that the accelerometer bias is not constant, it means it is necessary to estimate sensor bias within short time period (in this case 5 s is considered as one period). In the second example in Fig. 5, we could see that the data obtained from the GPS is not continuous since the GPS module sometimes loses track of satellite (the device is installed under the dashboard, and does not have a very clear view of the sky).

6.1.2. DCM filter for attitude angle estimation

Another step needed for trajectory estimation is to estimate the orientation of the probe vehicle, that is, the attitude of the vehicle in the Earth frame. The Direction Cosine Matrix (DCM) filter is commonly used for ground vehicle attitude estimation and control [31–33]. A DCM filter is ran on board the proposed platform for estimating the Euler angles: roll (φ), pitch (θ) and yaw (Ψ) of the vehicle. This complementary filter is based on Eq. (2).

$$\begin{cases} \varphi(t + \Delta t) = \lambda \cdot (\varphi(t) + g_x(t) \cdot \Delta t) + (1 - \lambda) \cdot \varphi_{ref}(t); \\ \theta(t + \Delta t) = \lambda \cdot (\theta(t) + g_y(t) \cdot \Delta t) + (1 - \lambda) \cdot \theta_{ref}(t); \\ \Psi(t + \Delta t) = \lambda \cdot (\Psi(t) + g_z(t) \cdot \Delta t) + (1 - \lambda) \cdot \Psi_{ref}(t); \end{cases} \quad (2)$$

In this equation, $g_x(t)$, $g_y(t)$, $g_z(t)$ are the rotation rates around the lateral, longitudinal and vertical axes of the vehicle, as measured by gyroscope at time t , and $\varphi_{ref}(t)$, $\theta_{ref}(t)$, $\Psi_{ref}(t)$ are the reference Euler angles, which can be obtained through data fusion of accelerometer and magnetometer measurements at time t (see [25] for details on the generation of these reference angles). λ is a dimensionless coefficient that trades off the current estimate of Euler angles with the corresponding measurement (reference value), similarly to a Kalman gain, which would be represented here by $1 - \lambda$. The choice of λ depends on the relative precision of the accelerometer and magnetometer, and of the gyroscopes, perfect gyroscope data would be associated with $\lambda = 1$. For this article, we use a value of $\lambda = 0.95$. The heading angle estimation results get from real tests are shown in Fig. 6. Based on the estimated Euler angle of the vehicle, the rotation matrix ($R_{s/g}$) transforming the ground coordinates into the sensor coordinates can be determined to correct of the acceleration measurement. If

$a_m(t)$ is the measurement data get from accelerometer, then the acceleration in device frame becomes $a_m(t) - R_{s/g}(t) \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix}$.

Thus the acceleration in vehicle frame ($a_v(t)$) can be obtained through Eq. (3).

$$a_v(t) = R_{s/v} \times \left(a_m(t) - R_{s/g}(t) \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} \right) \quad (3)$$

6.1.3. Trajectory estimation

Once the speed ($v(t)$) and heading angle ($\psi(t)$) of the vehicle are obtained, we can easily get the trajectory of the vehicle through the Eq. (4). Examples of real trajectory estimation results are shown in Fig. 7).

$$\begin{cases} X(t + \Delta t) = X(t) + v(t) \cdot \cos(\psi(t)) \\ Y(t + \Delta t) = Y(t) + v(t) \cdot \sin(\psi(t)) \end{cases} \quad (4)$$

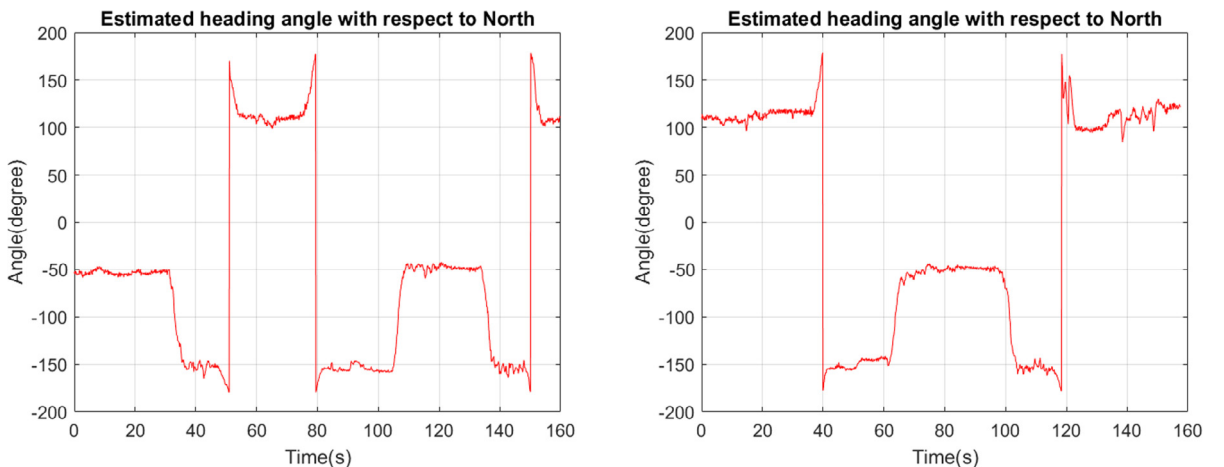


Fig. 6. Heading angle estimation results based on DCM filter in real tests.

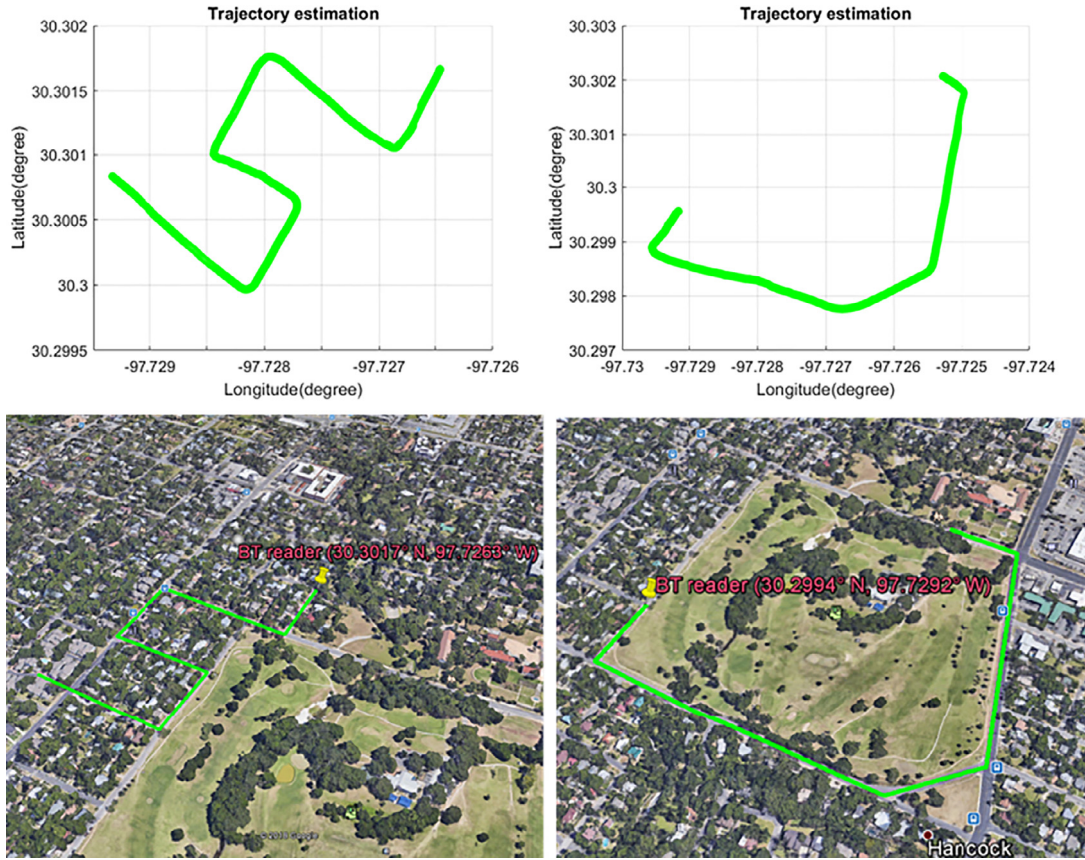


Fig. 7. Estimated trajectories and its path in real map.

6.2. Road surface condition monitoring

6.2.1. Pavement roughness and PSR

Pavement roughness is an important characteristic used to indicate the condition of road surface, which affects not only ride quality but also vehicle delay costs, fuel consumption and maintenance costs. Pavement roughness is typically quantified using some form of either Present Serviceability Rating (PSR), International Roughness Index (IRI) or other index with IRI being most prevalent [34,35]. In the present work, PSR is used as the index of road surface condition, which is defined as the judgment of an observer as to the current ability of a pavement to serve the traffic. The quantitative scale of PSR ranges from 5 (excellent) to 0 (essentially impassable). Since the PSR value is based on passenger interpretations of ride quality, it generally reflects road roughness because roughness largely determines ride quality.

Series of routes in the Austin, Texas with certain PSR values were chosen as the test routes for vehicle equipped with our sensing platform. Those routes are classified based on different levels of PSR and are illustrated with different colors, as showed in Fig. 8. The tests were conducted with the same type of vehicle (Ford F150) that was used to obtain the PSR value of these routes. Test vehicle and device connection inside the vehicle are shown in Fig. 9.

6.2.2. Road surface monitoring with the proposed platform

The main idea of road condition monitoring with our proposed platform is to detect the pavement roughness with the IMU data which includes acceleration rate. This is a continuous sensing approach that the data is continuously sampled from the probe vehicle itself. The main source of data used for this process is the vehicle's vertical acceleration. The solution is to explore the possible relationship between the vertical acceleration and the PSR so that the road surface condition can be estimated in real time based on this relation.

The idea of this article is to explore the relationship between predictor variables extracted from the vertical acceleration data and the PSR. To build the relationship between PSR and the acceleration rate, the first step is to sample the acceleration data (mainly the vertical acceleration rate) and separate them into different frequency components using Fast Fourier Transform. In this case, the vertical acceleration signal is sampled into six different bands which are 0.5–1.5 Hz, 1.5–2.5 Hz, 2.5–3.5 Hz, 3.5–4.5 Hz, 4.5–5.5 Hz, 5.5–6.5 Hz (as shown in Fig. 10). The 0–0.5 Hz band is excluded since it mainly corresponds to fluctuations in the slope of the road, and is not related to the pavement condition.

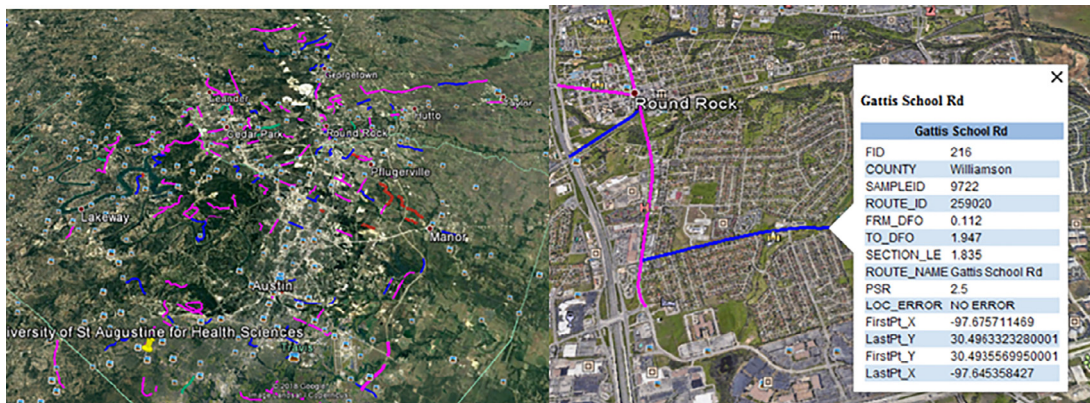


Fig. 8. Routes with different levels of PSR in the Austin Area and the specific route information Color meaning: $1 < \text{PSR} \leq 2$ (Red); $2 < \text{PSR} \leq 3$ (Blue); $3 < \text{PSR} \leq 4$ (Pink); $\text{PSR} = 4$ (Green).



Fig. 9. Test vehicle and device connection inside the vehicle.

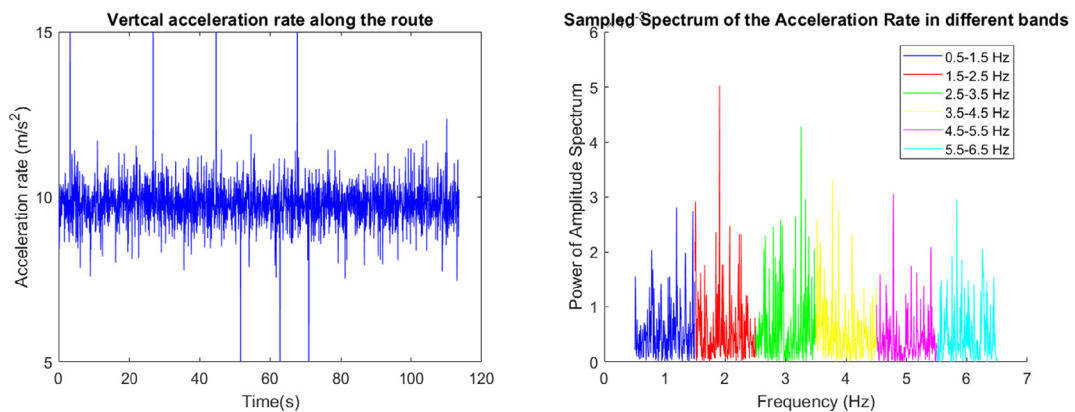


Fig. 10. Vertical acceleration rate along time (left) and the distribution of the sampled spectrum after FFT (right).

Once the frequency distribution of the sampled acceleration is obtained, the relationship between the data in different bands with PSR can be inferred. As running speed is a parameter that would affect the measured acceleration, the variations in the vehicle speed while driving the route are also considered. If a_z represents the average power of the vertical acceleration in a frequency band, the normalized $\frac{a_z}{v^2}$ can be related the PSR, where v stands for the average running speed along the route. During the test, we attempted to keep the same speed while running through one route, which is close to the speed limit of the route and also the same speed as when the PSR value is obtained.

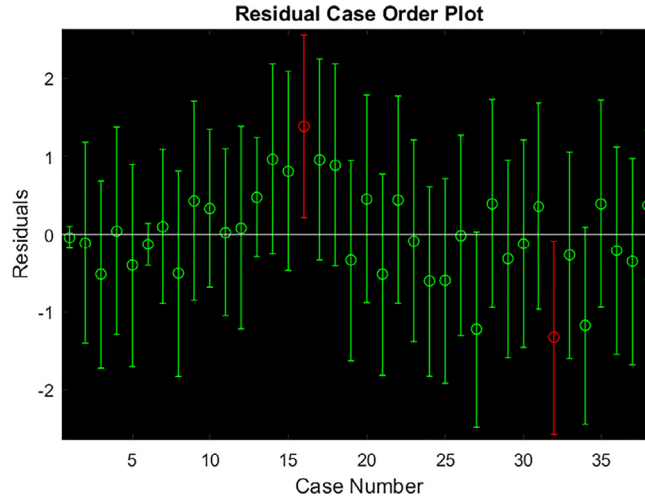


Fig. 11. Residual of linear regression.

After 45 different routes in Austin, Texas were tested and some outliers were excluded, 38 sets of data were left for exploring the possible relationship between PSR and vertical acceleration. To avoid over fitting, we performed a simple linear regression represented in Eq. (5). The residuals of this regression are plotted in Fig. 11.

$$PSR = \alpha + \alpha_1 \cdot x_1 + \alpha_2 \cdot x_2 + \alpha_3 \cdot x_3 + \alpha_4 \cdot x_4 + \alpha_5 \cdot x_5 + \alpha_6 \cdot x_6 \quad (5)$$

where

$$x_k = \frac{a_z(k)}{v^2}, \quad k = 1, 2, 3, 4, 5, 6; \quad (6)$$

$$\alpha = 3.0629, \quad \alpha_1 = -7729, \quad \alpha_2 = -4217.4, \quad \alpha_3 = 18657, \quad \alpha_4 = 11903, \quad \alpha_5 = -6737.3, \quad \alpha_6 = -18154.$$

In Eq. (6), the value k represents different bands: 1 refers to band 0.5–1.5 Hz, 2 refers to band 1.5–2.5 Hz, 3 refers to band 2.5–3.5 Hz, 4 refers to band 3.5–4.5 Hz, 5 refers to band 4.5–5.5 Hz, and 6 refers to band 5.5–6.5 Hz.

From the preliminary model we got, the PSR value shows an inverse growth relationship with the value of vehicle's vertical acceleration divided by average speed in the 0.5–1.5 Hz, 1.5–2.5 Hz, 4.5–5.5 Hz and 5.5–6.5 Hz bands. For these bands, the lower vertical acceleration rate, the better road surface condition is (smoother road surface). As can be seen from Fig. 11, the residuals are on the order of ± 1 , which may be related to several factors:

- Errors in calibration of the accelerometer
- Additional predictor variables that are not considered in this article (for example, rotation rate data)
- Subjectivity in the determination of the PSR by humans.

7. Conclusion and future work

This article introduces an open-source IMU/GPS device that can be used for traffic and road condition monitoring applications. The device is based on a 9 degree of freedom IMU (accelerometer, gyroscope and magnetometer), with a GPS, and Bluetooth communication. Example applications include trajectory estimation of road vehicles, and pavement condition monitoring. Future work will focus on improving the reliability of the system by adding a watchdog-based reset. Also, real time operating system (RTOS) will be used for better performance of the system for complex applications. It will also involve the use of a multi-rate fusion algorithm (for example based on Kalman filtering) for processing the sensor data.

Declaration of interests

None declare.

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